

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Electrical Bulk MTOs
Document Number	EA-BFPCL-GPMC-GGPL-FEED- ELE-LST-004
Document Revision	R01
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Originator	Noel Kari
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Issue Date	8 Jun 2022

Electrical Bulk MTOs

R01	08.06.22	Issued For Review	N.Kari	V.Ani	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

R01	08.06.22	Issued for Review	N.Kari	V.Ani			
Rev No	Date	Reason for Issue / Change	Prepared	Reviewed	TCE Approval	BFPCL Approval	Approver (BFPCL)

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

Table of Contents

1. GAS PROCESSING AND METHANOL COMPLEX
PROJECT FEED FOR GAS GATHERING PIPELINE
Electrical MTO

GAS PROCESSING AND METHANOL COMPLEX PROJECT FEED FOR GAS GATHERING PIPLINE Electrical MTO

S/No	Material Description	Units	Quantity	Contingency	Total Quantity	Cost/Price (NGN)		Manufacturer	Model	Remarks
						Unit Rate	Total Sum			
Cables (High Voltage & Low Voltage)										
1	3C x 70mm ² Power cable	mtrs	38500	500	39000	24,500.00	955,500,000.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 6.35/11kV	Amoured Underground Power Cable From AKABEL Power Plant
2	4C x 16mm ² Power cable	mtrs	150	23	175	4,000.00	698,000.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 0.6/1kV	From Step-down Transformer to Distribution board.
3	70mm2 Raychem Cable jointing Kit	pcs	77	12	91	12,000.00	1,086,600.00	NEXANS Kabelmetal or Nigerchin.	Cu/XLPE/PVC/SWA/PVC, 0.6/1kV	For Cable Jointing
Transformers, Substation Materials & Accessories										
4	200A, 11kV, SF6, RE2d, Ring Main Unit(RMU) with one incomers and two Outgoers.	pcs	4	0	4	4,500,000.00	18,000,000.00	ABB/Schneider Electric or approved equivalent		RMU to be installed at each Station including Akabel Power Station
5	100A, 400/230V AC, 50HZ, TPN+E, 5W,IP42, with 100A, incomers and Outgoers as detailed in the SLD.	pcs	3	0	3	450,000.00	1,350,000.00	ABB/Schneider Electric or approved equivalent		
6	32A, 400/230V AC, 50HZ, TPN+E, 5W,IP42, photocell Distribution board with 32A incomer and Outgoers as detailed in the lighting Schematic SLD.	pcs	3	0	3	450,000.00	1,350,000.00	ABB/Schneider Electric or approved equivalent		
7	5kVA Dual Redundant UPS including Battery Back-up for 80% load for 4hrs and dedicated UPS panel and all installation accessories	pcs	3	0	3	5,000,000.00	15,000,000.00	ABB/Schneider Electric or approved equivalent		
8	50kVA, 11/0.4 KV, 50Hz, Dyn11, KNAN Step - down, Modular transformer complete with mounting accessories.	pcs	3	0	3	4,700,000.00	14,100,000.00	ABB/Schneider Electric or approved equivalent		
9	50kVA, 0.4/11 KV, 50Hz, YnD11, KNAN Step - up, Modular transformer complete with mounting accessories.	pcs	1	0	1	4,700,000.00	4,700,000.00	ABB/Schneider Electric or approved equivalent		
	Installation, Testing & Commissioning									
10	Transformers , RMUs, UPS, Panels, installation, pre-commissioning & Commissioning	Lots	1	0	1		20,000,000.00			
11	Cable Trench Excavation & Cable Installation.	Lots	1	0	1		300,000,000.00			
12	Sub Total						1,331,784,600.00			
13	VAT (5%)									
14	Grand Total									
	NOTE 1. Ref single line diagram for connections									